Convergence of Fourier Partial Sums at Jump Discontinuities

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October 21, 2026

Abstract

This manuscript examines the pointwise behaviour of Fourier partial sums of a square wave. Away from the discontinuity, partial sums converge to the generating function; at the discontinuity, they overshoot by a fixed proportion that does not vanish as terms accumulate. Numerical evaluation for cutoffs of one to thirty-one terms shows the maximum rising towards the Wilbraham-Gibbs value $(2/\pi)\,\mathrm{Si}(\pi) \approx 1.1790$ while the normalized mean-square error decays from 0.189 to 0.007. The study is a publishing fixture: it verifies that abstracts, keywords, labeled equations, cross-references, a plotted figure, a results table, and structured references survive conversion into journal manuscripts, and it claims no new mathematics.

1 Introduction

Fourier's expansion of periodic functions into trigonometric series is the founding result of harmonic analysis [1]. For a function with a jump discontinuity, however, the convergence of the partial sums is subtle: observers recorded an unexplained overshoot near the jump long before it was analyzed [2], and the effect was later attributed to the series itself rather than to computational artifacts [3]. The same reference is cited again [3] to verify stable numbering under export.

A journal manuscript combines several kinds of information. Its abstract states the contribution, the body develops the model, and figures, tables, and references carry evidence that must survive conversion. This manuscript exercises that structure end to end; its numerical experiment is deliberately small so that export behaviour can be inspected independently of scientific novelty.[1]

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[1] All authors and institutions are illustrative. The sine integral $\mathrm{Si}(\pi)$ is evaluated numerically; the experiment is reproducible from the stated formulas alone.

2 Partial Sums and Pointwise Convergence

Consider the unit square wave with $f(x) = 1$ on $(0, \pi)$ and $f(x) = -1$ on $(-\pi, 0)$. Its K-term partial sum retains the first K nonzero odd harmonics, as defined in Equation 1:

$$S_K(x) = \frac{4}{\pi} \sum_{j=0}^{K-1} \frac{\sin\big((2j+1)x\big)}{2j+1}. \tag{1}$$

Two quantitative facts frame the experiment. First, the overshoot does not disappear: at the points approaching the jump, the partial sums converge to the value stated in Equation 2. Second, the energy of the residual decays to zero: Parseval's identity gives the normalized mean-square error in Equation 3.

$$\lim_{K\to\infty} S_K\Big(\frac{\pi}{K+1}\Big) = \frac{2}{\pi}\,\mathrm{Si}(\pi) \approx 1.1790. \tag{2}$$

$$E_K = 1 - \frac{8}{\pi^2} \sum_{j=0}^{K-1} \frac{1}{(2j+1)^2}. \tag{3}$$

3 Illustrative Results

Figure 1 shows the fifteen-term partial sum against the square wave. The curve tracks the function away from the jump and overshoots it near $x = \pi/16$, consistent with Equation 2.

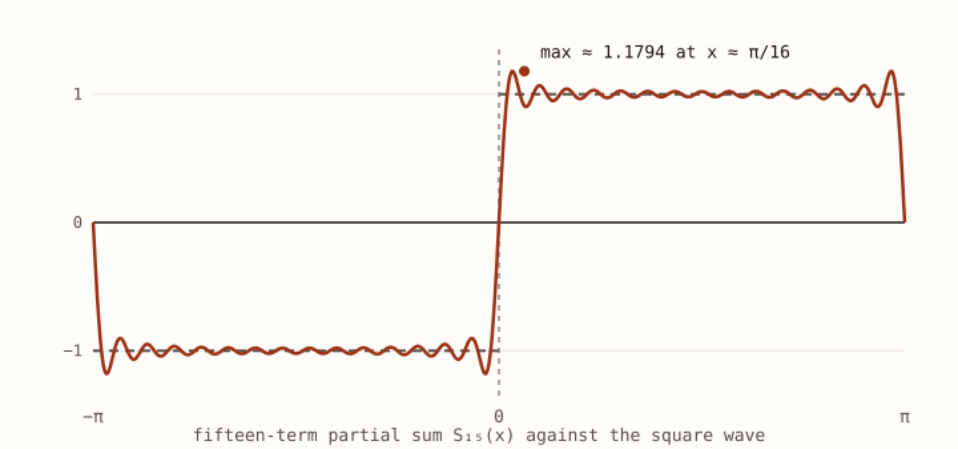


Figure 1: The fifteen-term partial sum S_{15} (solid) against the square wave (dashed). The maximum occurs near $x = \pi/16$ and approaches the Wilbraham-Gibbs value as terms accumulate.

Table 1 lists the measured maxima and mean-square errors. The maxima descend towards 1.1790 while the errors in Equation 3 fall below one percent by fifteen terms.

4 Conclusion

The experiment reproduces the classical picture: pointwise convergence away from the jump, a persistent overshoot of roughly nine percent of the jump near it, and mean-square convergence throughout. The manuscript is a fixture for the export pipeline rather than a claim of new results; its abstract, keywords, labeled equations, cross-referenced figure and table, footnote, and structured references are the payload under test.

References

1. Joseph Fourier, Theorie analytique de la chaleur (Analytical Theory of Heat) (1822)

2. Henry Wilbraham, On a certain periodic function (1848)

3. Josiah Willard Gibbs, Fourier's series (1899)

Table 1: Convergence of square-wave partial sums: the maximum approaches the Wilbraham-Gibbs value $(2/\pi)\,\mathrm{Si}(\pi) \approx 1.1790$ while the normalized mean-square error decays.

Terms K	**Maximum of** S_K	**Mean-square error** E_K
1	1.273240	0.189431
3	1.188357	0.066944
7	1.180684	0.028900
15	1.179350	0.013504
31	1.179066	0.006536